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An intuitionistic fuzzy deep neural network method for risk assessment of oil and gas pipelines

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ABSTRACT

Oil and gas transportation pipelines are complex systems inherently susceptible to failure, with severe consequences. An effective risk management system is required to ensure the safe and efficient operation of pipelines. However, traditional models for pipeline integrity risk assessment are often conservative and struggle with uncertainty and non-linear relationships in complex operational data, resulting in reduced predictive accuracy and ineffective decision-making. This study proposes a novel risk model to enhance pipeline risk assessment. The framework builds on the Muhlbauer model and API RP 581 as baseline methodologies, integrating with an intuitionistic fuzzy deep neural network. Intuitionistic fuzzy sets explicitly represent uncertainty through membership and non-membership degrees, whereas deep neural networks capture complex non-linear patterns in risk-related data. The performance of the proposed framework was validated on real-world pipelines and compared with traditional approaches. The results demonstrate that the hybrid model improved predictive accuracy, provided more consistent risk estimation, and effectively managed data uncertainty and non-linearity. The proposed methodology offers an advanced decision-support tool for asset integrity management, enabling more effective, risk-informed pipeline inspection and maintenance strategies.

1. Introduction

Failure of equipment in energy facilities poses a major hazard that can impact personnel safety, economic losses, environmental pollution, and the company's reputation. An oil and gas facility, as a key infrastructure of the energy sector, involves complex operations; in the event of a failure, it can cause major consequences, such as loss of containment, facility damage, and unplanned shutdowns (W. Wu et al., 2013). Therefore, the risk of failure and associated financial losses must be addressed in advance. Integrity management (IM) to prevent equipment failure in the petroleum industry is critical for maintaining the global energy supply (Oropallo et al., 2024). IM implementation provides various strategic benefits, such as enhancing the service life of objects by enabling efficient integrity inspection and renewal. Inspection and maintenance operations can also be prioritized to achieve optimal resource use, minimum downtime, and drastically lower repair and lifecycle costs (Kang et al., 2024).

The IM program focuses on major accidents involving pressurized equipment. This program was developed and systematically implemented to prevent equipment failures, reduce associated risks, and ensure safe operation throughout its service life (Wahono et al., 2025). The most common factors in equipment failure in pipeline operations are corrosion-related material degradation, third-party interference, and exposure to natural hazards. These are significant determinants of pipeline integrity, safety, operational breakdowns, and costs (Abubakirov et al., 2020; Eastvedt et al., 2022; Mahmood et al., 2024). The Macondo well blowout incident, which impacted safety, financial, and environmental aspects and caused a leading oil and gas company to file for bankruptcy, is one of the best examples of the lack of implementation of asset integrity (Chandima Ratnayake and Markeset, 2012).

Risk-based decision-making improves safety performance and achieves an optimal cost of inspection and maintenance. From an asset integrity perspective, a risk-based approach is founded on the premise that most high-risk equipment is concentrated in a small portion of the plant or facilities. An organization may allocate its spending to high-risk rather than lower-risk equipment to optimize operations and capital expenses (Bhatia et al., 2019). The risk-based concept in inspection and maintenance addresses challenges and provides solutions to reduce

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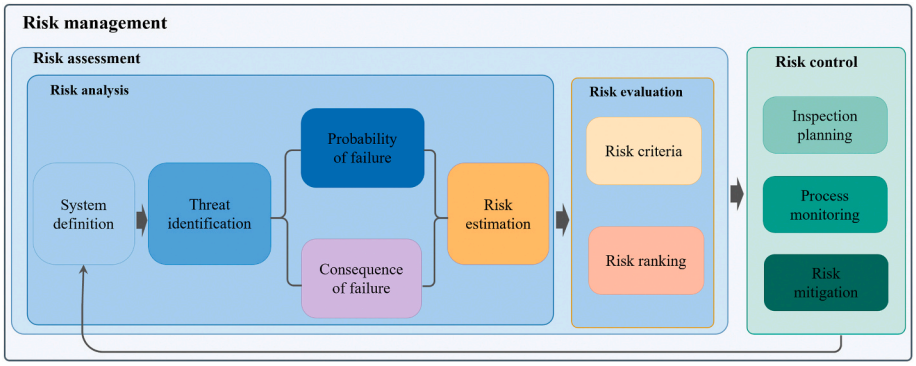


Fig. 1. Risk management process adapted from API RP 580 (2023).

operating costs and asset impairment in the global economy, especially in the petroleum industry (Aditiyawarman et al., 2023). Cost reduction is not the main driver of risk-based inspection (RBI) approaches; however, cost optimization is possible through more efficient cost reallocation (Oropallo et al., 2024).

Successful risk management is supported by accurate risk estimation and program effectiveness. Risk management activities involve risk assessment and control, with the analysis supported by a risk model. Risk analysis is essential for preparing safety strategies to prevent incidents. Over the past few decades, traditional methods, such as fault trees, event trees, bow-tie diagrams, and sequence-of-event diagrams, have been proposed as risk models in previous studies (Li et al., 2022). Some enhancement models, such as Bayesian networks, have been used to identify pipeline failures by predicting leakage or rupture events and to inform inspection strategies (Breton et al., 2010; Luque J, 2019). Fuzzy logic and its extensions have been used for risk analysis in the RBI program, providing a clear process that leads to more accurate results (Jamshidi et al., 2013; Sadeghi et al., 2025). Artificial intelligence (AI) was used to improve the quality of traditional RBI assessment outputs by minimizing variability and improving both accuracy and precision (Rachman and Ratnayake, 2019; Taufik et al., 2023).

The role of artificial intelligence in the energy sector has become a core enabler for efficiency, safety, and sustainability. In the oil and gas industry, it is one of the fastest to adopt new technologies to manage its business, where AI plays an important role in monitoring assets, managing risks, and making critical decisions. The main targets of AI (or any other digitalization) are efficiency, process acceleration, and risk reduction (Koroteev and Tekic, 2021). AI has supported the industry by providing critical information through organized, transparent, and accessible databases, thereby enhancing decision-making for operations and investments (Z. Chen et al., 2025). The use of AI in risk analysis is an input-algorithm-output process, where the description of the risk is the main input and produces potential outputs, with key elements in the form of (risk sources, events/scenarios, and effects), uncertainty measures, and knowledge bases built from the previous two elements (Stødle et al., 2024). Specific use of AI in pipeline integrity management (PIM) to support improved analysis and the potential of data-driven approaches for the most critical threats, such as third-party damage and external corrosion (Bastek et al., 2026).

Despite providing considerable benefits to the risk assessment process, the conventional and existing enhancement methods reveal a notable gap in prior research, particularly in the absence of a hybrid research design that integrates uncertainty modeling into pipeline RBI within the oil and gas industry. This drawback has had a detrimental effect on operators and stakeholders. Existing AI methods do not fully account for uncertainty; AI models can sometimes produce false results, and their outputs are difficult for decision-makers to interpret (Stødle et al., 2024). Moreover, combining AI and fuzzy reasoning to enhance risk assessment systems would create new opportunities to address problems in the petroleum industry (Sadeghi et al., 2025).

This paper attempts to address the aforementioned challenges by increasing the complexity of systems and the potential consequences of failure, underscoring the need for robust risk modeling that explicitly accounts for uncertainty as a key priority to improve analysis accuracy and reduce conservatism in decision-making. Most previously published studies used probability and consequences to assess pipeline risk, although these parameters cannot account for all hazard aspects, including uncertainties. The system's uncertainty and intrinsic complexity cannot be addressed by Aristotelian logic (Raeihagh et al., 2020). There is still a long way to go before it is reached due to uncertainties and inconsistencies in the data and information gathered, modeling, and material characteristics under damage (Sadeghi et al., 2024). Risk assessment accuracy is important to the success of an asset integrity program in creating inspection and maintenance plans. The accuracy of low-risk assessment results leads to less frequent inspections of high-risk equipment that poses a failure risk, and increasing the number of inspections of low-risk equipment would be more expensive (Rachman and Ratnayake, 2019).

A risk assessment method based on an intuitionistic fuzzy approach yields better results than the traditional method (Sadeghi et al., 2025). This methodology provides higher accuracy and more realistic, reliable risk estimation, enhancing asset and operational protection while reducing inspection frequency and maintenance costs in the oil and gas industry (Zhang et al., 2024). AI offers significant benefits for risk prediction by effectively and accurately evaluating complex risk data (Taufik et al., 2023). AI has strong predictive capabilities but struggles to handle uncertainty. A novel approach that combines deep learning and fuzzy systems was recently investigated (Y. Zheng et al., 2022). A hybrid method that integrates deep neuro-fuzzy systems addresses the complexity of high-dimensional data and achieves better results in real-world classification challenges (Talpur et al., 2022).

This study proposes a new risk assessment model to address the limitations of traditional, classical fuzzy, and neuro-fuzzy models in managing the uncertainty and complexity of risk-related information. This study aimed to develop a novel hybrid intuitionistic fuzzy deep neural network model for pipeline risk assessment. Intuitionistic fuzzy sets are specifically used to effectively capture uncertainty, leading to improved accuracy and greater interpretability of assessment results. This method extends classical fuzzy set theory by explicitly integrating both membership and non-membership degrees, thereby representing a more complex and flexible model of uncertainty and hesitancy than fuzzy sets. Deep neural networks (DNNs) extend the ability to model complex non-linear relationships of risk-related data. The integration of intuitionistic fuzzy deep neural networks enables the model to handle both uncertainty and complex non-linear relationships, overcoming the limitations of traditional risk assessment methods by incorporating linguistic uncertainty, capturing input-output relationships, and considering the relative importance of risk factors, resulting in more accurate and realistic risk predictions. To demonstrate the detailed processes of the proposed method, a case study of the onshore oil and gas pipeline

system is implemented. Then, comparative analyses are performed to verify the improvement of the proposed model for pipeline risk assessment.

The structure of this paper is as follows: the second section provides a brief literature review of related work on pipeline risk assessment, including the application of fuzzy systems and deep learning. The third section will cover the research methodology, which describes the conceptual framework and research design. The fourth section presents the significant results and analysis of this study. Finally, the paper presents the conclusion and a suggestion for future work in the fifth section.

2. Related work

The risk management process involves strategizing and implementing appropriate risk-reduction measures in ongoing pipeline activity management, followed by continuous improvement. Risk management strategies allow operators to allocate limited resources to maintain the asset based on the assessed risk of failure, thereby reducing the probability of an accident (Singh and Pokhrel, 2018). Fig. 1 shows the flowchart of the risk management process. The steps of the risk assessment process comprise damage mechanism identification, risk analysis, and evaluation to enhance the reliability of a pipeline system (Khan et al., 2021). Risk analysis involves threat identification and evaluation of the probability of failure (PoF) and consequences of failure (CoF) as a basis for risk estimation. Furthermore, risk evaluation is performed by comparing it with the criteria based on an established standard. Risk control reduces risk by preventing, mitigating, and monitoring hazards and their associated consequences, and by effectively communicating risk management strategies (ISO, 2021).

The risk modeling framework selected during operational pipeline risk management determines the effectiveness of the risk management process. The model has inputs, outputs, and an algorithm to assess its capacity to support decision-making in pipeline risk management and to comply with regulations. An effective risk model should integrate a range of causative elements and their interactions, rather than focusing solely on a single major cause of failure (Abubakirov et al., 2020; Zhao et al., 2025). Risk models can be divided into basic qualitative and relative assessment/indexing models, and more advanced quantitative and probabilistic models (Animah and Shafiee, 2020; Woldesellasse and Tesfamariam, 2023).

Qualitative and relative/index risk models are generally applied in pipeline risk assessment methodologies for baseline integrity assessment. This method was chosen for its straightforward, fast approach, based on expert appraisal rather than quantitative or computer calculations (Wahono et al., 2025). A qualitative approach using the relative risk score method offers time efficiency and cost-effectiveness. The advantages of this method include its ability to incorporate multiple parameters to forecast the potential for leaks and ruptures (Vanitha et al., 2023). The weakness of the qualitative method as an initial risk screening and prioritization tool is that its results are expressed as relative risks, are non-proportional, and lack a standard risk unit. Therefore, it applies to small operating units but requires additional data and analysis to support comprehensive decision-making.

In contrast, a quantitative system model uses computational and mathematical models to estimate the aggregate risk in consistent units. The quantitative approach is data-driven and susceptible to overlooked inspection data or errors under more complex operating conditions (Fang et al., 2019). A layer-of-protection analysis, in which qualitative severity assessments are replaced with more accurate and effective risk evaluations, is one example of a quantitative approach (Yan and Xu, 2018). These models are designed to estimate risk before and after risk reduction because a quantitative system model represents the model's input physical and logical relationships. Risk reduction should be based on a cost-benefit analysis to inform risk management decisions. When detailed, accurate data are available, quantitative studies are optimal but may be time-consuming and costly. However, collecting all data and information for analysis is difficult due to operational complexity (Yin et al., 2021).

A probabilistic model supports all decision types by incorporating uncertainty in model inputs through probability distributions, enabling the generation of output distributions consistent with the operator's risk tolerance (Singh and Markeset, 2014). An integrity assessment interval is preferably identified using a probabilistic model with input and output distributions, as this allows estimation of the uncertainty in the predicted failure probability given the actual integrity assessment findings. Probabilistic techniques (e.g., reliability and sensitivity analyses) are important for evaluating risk in decision-making under uncertainty (Alsulieman et al., 2024; Lu et al., 2021). Probability distributions can be used to model uncertainty in tool tolerances and to propagate risks (such as corrosion growth, excavation damage, and equipment reliability) into the risk model, producing a more realistic distribution of failure probability. Probabilistic models are superior at handling stochastic variability, but their accuracy diminishes significantly when faced with limited or sparse data, often resulting in unreliable predictions. They may often be ineffective at capturing epistemic uncertainty (uncertainty due to insufficient knowledge) typical of real-world pipeline data, thereby yielding inaccurate estimates of failure probabilities (W. Wu et al., 2019).

The main goal of any risk model is to estimate the actual risk, so incorporating uncertainty into the model is important. Variations in the input model will have an impact on the results, and different parameters have different influences on the results. To address these challenges, several uncertainty modeling theories have been developed. Probability theory, including Bayesian Networks, Monte Carlo Simulation, and reliability analysis, has been widely adopted to address uncertainty and failure probability in risk assessment (Aven, 2020; Ghasemi Pirbalouti et al., 2023; Jiang et al., 2026; W. Wu et al., 2019; Xie and Tian, 2018). However, this approach has a limitation because it requires sufficient data records and a predefined probability distribution, which are often unavailable in pipeline integrity applications. Dempster–Shafer (D–S) evidence theory extends probabilistic reasoning to represent incomplete and conflicting information without requiring prior probabilities. By using belief and plausibility functions, it explicitly captures ignorance and combines evidence from multiple sources, making it suitable for risk assessment under uncertainty (Jing et al., 2026; Xu et al., 2022; Zhu and Xiao, 2021). However, its computational complexity increases when dealing with large numbers of hypotheses and conflicting evidence. The Cloud Model (CM) is an uncertainty modeling approach that simultaneously captures fuzziness and randomness, enabling a more comprehensive representation of uncertainty than traditional fuzzy set theory (Y. Chen et al., 2022; Gao et al., 2022; S. Wu et al., 2026). Despite its advantages, existing CM applications often rely on fixed-weight schemes, simplified expert judgment, and incomplete uncertainty transformations, thereby reducing the effectiveness of uncertainty representation in risk assessment. Among these uncertainty theories, Fuzzy Set Theory (FST) has been widely applied in risk assessment because it can represent vagueness and imprecision through linguistic variables and membership functions while maintaining computational simplicity and interpretability (Markowski and Mannan, 2009; Zadeh, 1965). Recent studies have further demonstrated the effectiveness of a fuzzy-based approach for pipeline risk assessment (Singh and Markeset, 2009). developed a fuzzy framework for risk-based inspection planning and calculating the estimate of corrosion rate, highlighting flexibility and easy modification of the input and output variables, a better transparency model, and integrating expert knowledge and objective data (Jamshidi et al., 2013). developed a fuzzy inference system for pipeline risk assessment, highlighting that the model provides more accurate, precise, and certain results (Osman and Shehadeh, 2022). developed a fuzzy-clustering framework for interstate pipeline risk assessment, highlighting that the model is proven to be an efficient model for pipeline risk assessment.

To improve the representation of uncertainty, several extensions of

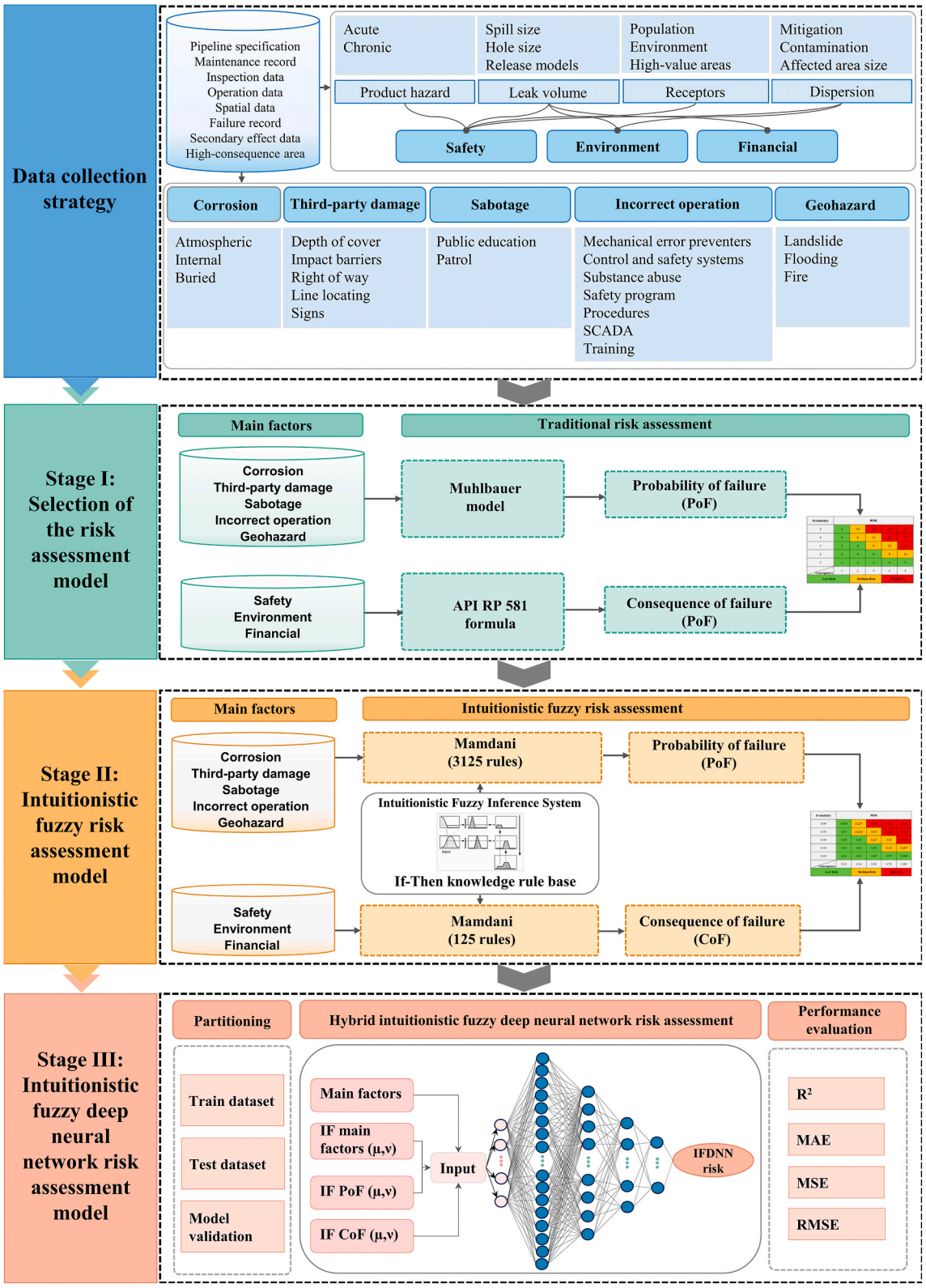


Fig. 2. Proposed pipeline risk assessment modeling framework.

classical fuzzy theory have been proposed. Interval Type-2 Fuzzy Sets (IT2FS) model uncertainty by interval-valued membership functions so that they can treat high-uncertainty data. IT2FS has been applied in the risk assessment and corrosion failure probability of oil and gas pipelines (Jana et al., 2017; Liu et al., 2024; Xu et al., 2022). However, it will be confronted with increased computational complexity and reduced interpretability. More recently, T-Spherical Fuzzy Sets (T-SFS) represent an advancement of fuzzy set theory, characterized by their lack of strict limitations (Huang et al., 2023; L. Zheng et al., 2025). Despite their advantages, greater parameterization increases model complexity, potentially limiting practical implementation. Alternatively, Intuitionistic Fuzzy Sets (IFS) extend classical FST by explicitly integrating both

membership and non-membership degrees, thereby representing a more complex and flexible model of uncertainty and hesitancy than fuzzy sets (Atanassov, 1986). Three-component structures (membership, non-membership, and hesitation degrees) enable the intuitionistic fuzzy inference system (IFIS) to effectively capture incomplete, imprecise, and conflicting information, resulting in more representative and stable assessment outcomes. Experimental findings show that integrating IFIS with a deep learning model significantly enhances overall performance (Mukhlash et al., 2026). In the pipeline domain (Yu et al., 2022), proposed an intuitionistic fuzzy probabilistic Petri net method for risk assessment of submarine pipeline leakage failure, while (Sadeghi et al., 2024, 2025) developed an intuitionistic fuzzy approach in risk-based inspection with the case study of a steel tank and a pressure vessel, respectively. Existing research consistently demonstrates that IFS effectively captures uncertainty, leading to improved accuracy and greater interpretability of assessment results.

The increasing availability of inspection records, operational data, and historical failure databases has accelerated the adoption of machine learning (ML) and deep learning (DL) techniques for pipeline risk assessment. Traditional risk models require predefined rules and expert judgment, while ML algorithms can automatically identify complex non-linear relationships among risk factors, demonstrating strong performance in failure assessment, risk analysis, and risk-based inspection screening and assessment (Hu et al., 2025; Li et al., 2025; Rachman and Ratnayake, 2019; Taufik et al., 2023). Ensemble machine learning models, particularly Random Forest and AdaBoost, have demonstrated strong capability in predicting gas pipeline failures, achieving high accuracy while identifying critical influencing factors such as cathodic protection, pipe characteristics, and failure-related conditions, thereby supporting data-driven pipeline risk assessment and reliability management. The lack of accurate and complete pipeline data introduces uncertainty in evaluating site-specific failure factors, limiting the reliability and generalizability of risk assessment outcomes (Xiao and Xiong, 2025). Hybrid-AI-based risk assessment models have been proposed to address the limitations of conventional expert-based approaches in pipeline risk evaluation. By integrating the analytic hierarchy process (AHP) with a dual-hidden-layer adaptive regularized neural network (DHARN) model, enhanced accuracy and generalization capability were achieved, demonstrating its effectiveness for quantitative risk assessment of oil and gas pipelines under complex operating conditions (Qu et al., 2026). A comprehensive review of machine learning applications in risk assessment (Hegde and Rokseth, 2020) found that artificial neural networks (ANNs) are the most widely used machine learning technique in engineering risk assessment due to their ability to model complex non-linear relationships. Despite their potential, artificial intelligence (AI)-based risk assessments often struggle with uncertainty quantification, interpretability, and the reliability of their prediction, limiting their support for risk-informed decision-making (Stødle et al., 2024). To address these challenges, integrate FST-AI approaches to simultaneously handle uncertainty and complex non-linear relationships, improving prediction accuracy and decision performance compared with standalone methods (Raeihagh et al., 2020). developed a risk assessment of an onshore pipeline using ANN and a fuzzy inference system, highlighting that the model overcomes the limitations of traditional risk assessment methods by incorporating linguistic uncertainty, capturing input-output relationships, and considering the relative importance of risk factors, resulting in more accurate and realistic risk predictions (Yin et al., 2021). developed a criticality evaluation of oil and gas pipelines based on fuzzy logic and machine learning, highlighting that the framework provides automated evaluation, uncertainty handling, and enhanced prediction capability.

Despite previous research using fuzzy sets and artificial intelligence to evaluate risk, a major challenge remains: fully modeling and eliminating epistemic uncertainty, given the subjective and incomplete nature of pipeline integrity information. Existing models often oversimplify this uncertainty, as seen in probabilistic models with limited datasets, or lack the adaptive learning capabilities necessary to fully leverage diverse data sources, as is the case with conventional fuzzy logic. This study seeks to address this ongoing issue by introducing an innovative research methodology that combines an intuitionistic fuzzy set with a deep learning framework, providing a more sophisticated and accurate approach to managing epistemic uncertainty in pipeline risk assessment, a critical challenge arising from incomplete data, subjective expert judgments, and modeling simplifications that traditional methods often fail to quantify or mitigate adequately.

3. Methodology

Three principal stages structure the research methodology for pipeline risk assessment. The first stage involved selecting a risk model from established methodologies and considering the identified pipeline risk factors. In the second stage, the pipeline risk value was modeled as an intuitionistic fuzzy set using a sequential process comprising fuzzification, an intuitionistic fuzzy inference system, and defuzzification for each of the main factors. In the last stage, the intuitionistic fuzzy outputs are used as inputs and trained with deep neural networks to predict risk. The IFDNN risk assessment model was constructed using a Python-based deep learning framework (TensorFlow–Keras). The research method of this study follows the stages illustrated in Fig. 2.

3.1. Stage I: Selection of the risk assessment model

The Kent Muhlbauer model and API RP 581 were selected as the baseline framework for this study because of their ability to systematically represent probability and consequence factors in pipeline risk assessment (API RP 581, 2025; Muhlbauer, 2015). These methods identify all the main and sub-factors of pipeline risk to determine the PoF and CoF. In addition, the risk analysis is conducted using 31 sub-factors and 8 main factors, as illustrated in Fig. 2. The risk is evaluated based on these factors, and the PoF is assessed using the Muhlbauer model, which considers variables of exposure, mitigation, and resistance. On the other hand, failure outcomes, such as product hazards, leak volume, receptors, and dispersion, were categorized into safety, environmental, and financial outcomes using the API RP 581 formula.

In this study, failure was defined as a leak or rupture. An asset is in a state of failure if a credible failure mechanism occurs and the action taken to prevent this failure has not succeeded, or if the structural or operational reserve is insufficient to stop the degradation process. An exposure is a force or mechanism (without mitigation) that can cause failure if the resistance is insufficient. The next measures are the types and effectiveness of every mitigation measure, which aim to prevent or reduce exposure. Finally, a measure or estimate is needed to express the component's ability to withstand an exposure force without failure if the exposure reaches the component. Two types of failure mechanisms are time-dependent, i.e., corrosion (C), and time-independent, i.e., third-party damage (TPD), sabotage (SB), incorrect operation (IO), and geohazard (G) (Muhlbauer and Murray, 2024). The PoF is evaluated against the criteria and categorized on a scale of 1 to 5.

The consequences for safety (SF) are evaluated based on the potential impacts of fire and toxic exposure, with population density as the primary receptor. The severity of the damage caused by the released fluid affects the financial consequences (F). In the context of environmental (E) impacts, consequences are evaluated by examining the extent of the failure, as reflected in the spill volume and associated cleanup costs. Subsequently, the consequences are calculated and measured against established criteria, and a classification is assigned on a scale of 1 to 5. Finally, the overall risk is quantified by integrating the PoF with its consequences and is categorized as low, medium, or high (Fig. 2).

Table 1
A triangular intuitionistic fuzzy number represents the input and output variables.

Main factors	Very low (VL)	Low (L)	Moderate (M)	High (H)	Very high (VH)
Corrosion (C)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Third-party damage (TPD)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Sabotage (SB)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Incorrect operation (IO)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Geohazard (G)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Probability of failure (PoF)	(0.00, 0.01, 0.15; 0.95, 0.02)	(0.11, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.47, 0.60; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	0.70, 0.85, 1.00; 0.95, 0.02)
Safety (SF)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Environment (E)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Financial (F)	(0.0, 0.01, 0.15; 0.95, 0.02)	(0.12, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.50, 0.65; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	(0.70, 0.90, 1.00; 0.95, 0.02)
Consequence of failure (CoF)	(0.00, 0.01, 0.15; 0.95, 0.02)	(0.11, 0.25, 0.40; 0.85, 0.08)	(0.35, 0.47, 0.60; 0.85, 0.06)	(0.55, 0.70, 0.85; 0.90, 0.04)	0.70, 0.85, 1.00; 0.95, 0.02)

3.2. Stage II: intuitionistic fuzzy risk assessment model

The intuitionistic fuzzy risk assessment approach integrates traditional risk assessment principles with intuitionistic fuzzy sets to enhance the effectiveness and reliability of risk management, particularly in handling uncertainty and imprecision in incomplete and imprecise data. The Intuitionistic Fuzzy Inference System (IFIS) framework comprises three phases: fuzzification, a fuzzy inference mechanism that follows an inference engine and a rule base, and defuzzification, as depicted in Fig. 2 (Velu and Paulraj, 2025). The key components needed for these three stages are included in the intuitionistic fuzzy knowledge base. The PoF and CoF are transformed into intuitionistic fuzzy representations via a triangular intuitionistic fuzzy number-based fuzzification process, as presented in Table 1. Simultaneously, defuzzification is performed using Eq. (9). Integrating the PoF with the CoF yields the overall risk, which is subsequently categorized into risk levels (low, medium, and high), as illustrated in Fig. 6.

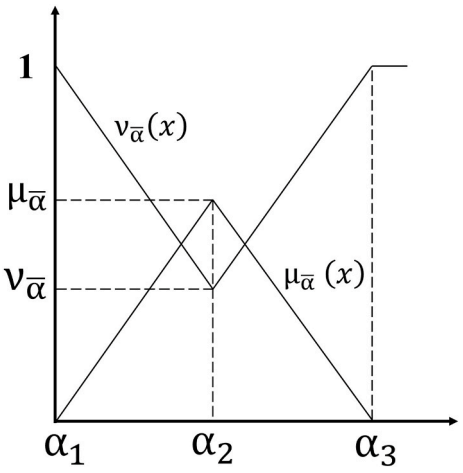


Fig. 3. The triangular intuitionistic fuzzy number.

Definition 1. (Atanassov, 1986) An IFS A on the universe of discourse E is represented as an object of the form $A = \{\langle (x, \mu_A(x), \nu_A(x) | x \in E)\}$, where μ_A: E → [0,1] and ν_A: E → [0,1] represents the membership and non-membership degrees of the element x∈E, respectively. These functions must satisfy the following condition: $0 \le \mu_A(x) + \nu_A(x) \le 1$. The degree of hesitation or uncertainty for each intuitionistic fuzzy set A defined on E is given by $\pi_A(x) = 1 - [\mu_A(x) + \nu_A(x)]$, where for all $x \in E$, $0 \le \pi_A(x) \le 1$.

Definition 2. (Velu and Paulraj, 2025) Let U ≠ Ø, V ≠ Ø, and A = (μ_A, ν_A), B = (μ_B, ν_B) are two IFS's of U and V, respectively. The Cartesian product A × B of U× V is defined as A× B = $(\mu_{A\times B}, \nu_{A\times B})$ where $\mu_{A\times B}(u,v) = \min\{\mu_A(u), \mu_B(v)\}$ and $\nu_{A\times B}(u,v) = \max\{\nu_A(u), \nu_B(v)\}, \forall u \in U, v \in V$.

Definition 3. (Sadeghi et al., 2025) A triangular intuitionistic fuzzy number, denoted as TIFN $\overline{\alpha}$ is defined by the following membership and non-membership functions, given by Eqs. (1) and (2).

$$\mu_{\overline{\alpha}}(x) = \begin{cases} (x-\alpha_1)\mu_{\overline{\alpha}}/(\alpha_2-\alpha_1) & \text{if } \alpha_1 \le x < \alpha_2 \\ \mu_{\overline{\alpha}} & \text{if } x = \alpha_2 \\ (\alpha_3 - x)\mu_{\overline{\alpha}}/(\alpha_3-\alpha_2) & \text{if } \alpha_2 < x \le \alpha_3 \\ 0 & \text{if } x < \alpha_1 \text{ or } x > \alpha_3 \end{cases} \quad (1)$$

and

$$\nu_{\overline{\alpha}}(x) = \begin{cases} [\alpha_2 - x + \nu_{\overline{\alpha}}(x-\alpha_1)]/(\alpha_2-\alpha_1) & \text{if } \alpha_1 \le x < \alpha_2 \\ \nu_{\overline{\alpha}} & \text{if } x = \alpha_2 \\ [x - \alpha_2 + \nu_{\overline{\alpha}}(\alpha_3 - x)]/(\alpha_3-\alpha_2) & \text{if } \alpha_2 < x \le \alpha_3 \\ 1 & \text{if } x < \alpha_1 \text{ or } x > \alpha_3. \end{cases} \quad (2)$$

If $\alpha_1 \ge 0$ and at least one of α_1, α_2, and α_3 is non-zero, then *TIFN* $\overline{\alpha}$ is positive and denoted as $\overline{\alpha} > 0$. Conversely, if $\alpha_3 \le 0$ and one of α_1, α_2, and α_3 is non-zero, $\overline{\alpha}$ is negative and denoted as $\overline{\alpha} < 0$.

3.2.1. Intuitionistic fuzzification

Fuzzification is the process of transforming crisp numerical values into grades of membership and non-membership for linguistic intuitionistic fuzzy sets. Association of each linguistic term with the grade is performed through the membership and non-membership functions. A triangular intuitionistic fuzzy number (TIFN) $\overline{\alpha}$ is an intuitionistic fuzzy set defined on the real number set R and is denoted as TIFN $\overline{\alpha} = \langle(\alpha_1, \alpha_2, \alpha_3; \mu_{\overline{\alpha}}, \nu_{\overline{\alpha}})\rangle$. Here, $\mu_{\overline{\alpha}}$ and $\nu_{\overline{\alpha}}$ denote the minimum membership degree and maximum non-membership degree, respectively, subject to condition $0 \le \mu_{\overline{\alpha}} \le 1$, $0 \le \nu_{\overline{\alpha}} \le 1$, and $0 \le \mu_{\overline{\alpha}} + \nu_{\overline{\alpha}} \le 1$. The graphical

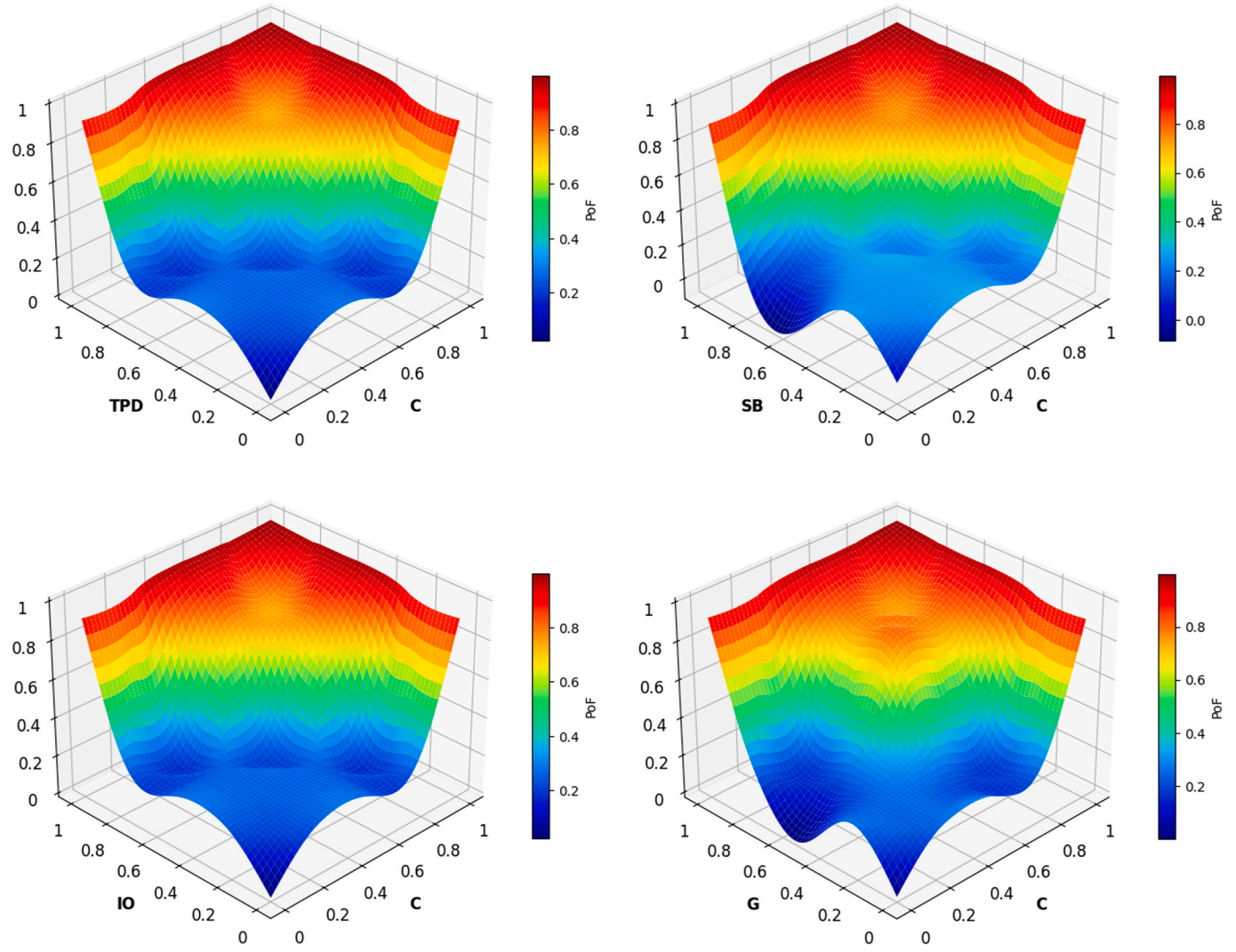


Fig. 4. Control surface of the effect of two input factors (C, TPD, SB, IO, or G) on PoF while other factors are constant.

representation of the triangular intuitionistic fuzzy number $\bar{\alpha}$ is illustrated in Fig. 3. All input variables and PoF and CoF as outputs applied in modeling are fuzzified with triangular intuitionistic fuzzy numbers with normalized values [0,1] by Eqs. (3) and (4). In this study, the triangular function was used because of its low computational cost and simple construction (Yin et al., 2021). The membership and non-membership degrees for the fuzzification process are numerically defined in Table 1 of the TIFN.

$$x_i^* = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{3}$$

$$y_i^* = \frac{y_i - y_{min}}{y_{max} - y_{min}} \tag{4}$$

where x^*_i (x_i) and y^*_i (yi) are the normalized values of the ith input and output, respectively. The terms $x_{\min}$ ($x_{\max}$) and $y_{\min}$ ($y_{\max}$) are the minimum (maximum) values of input and output variables, respectively.

3.2.2. Intuitionistic fuzzy inference system

An intuitionistic fuzzy inference system (IFIS) employs fuzzy If-Then rules with the Mamdani method to construct a connection from fuzzy inputs to outputs, following the principles of IFIS rule composition. Unlike the Tsukamoto FIS and the Sugeno fuzzy model, which produce constant or linear output values, the Mamdani method produces fuzzy output values (Raeihagh et al., 2020).

Consider a multi-input single-output (MISO) intuitionistic fuzzy inference system with two input variables x and y, and one output variable w. Let $A_i = (\mu_{A_i}, \nu_{A_i})$ and $B_i = (\mu_{B_i}, \nu_{B_i})$ denote intuitionistic fuzzy sets corresponding to the inputs, and $C_i = (\mu_{C_i}, \nu_{C_i})$ be the intuitionistic fuzzy sets assigned to the output, for the ith rule. The Mamdani algorithm in the most general If-Then rule form is expressed by Eq. (5) as follows (Velu and Paulraj, 2025):

I_i: If x is A_i and y is B_i Then w is C_i (for i = 1, 2, 3, …, m). (5)

Within the intuitionistic fuzzy framework, each rule I_i is represented by a membership degree μ_{I_i} and a non-membership degree ν_{I_i}, defined over the universe $U \times V \times W$.

Several fuzzy composition methods were used to describe the Mamdani fuzzy model. This study employs the most commonly used composition, the max–min composition (Jamshidi et al., 2013). This process is extended to the intuitionistic fuzzy framework, calculating the membership and non-membership functions. Thus, the degree of membership of the output is obtained using the sup-min operator, whereas the degree of non-membership is obtained using the inf-max operator and can be expressed by Eq. (6) as follows (Velu and Paulraj, 2025):

$$\begin{aligned} \mu_{I_i}(u,v,w) &= min\{\mu_{A_i}(u), \mu_{B_i}(v), \mu_{C_i}(w)\}, \nu_{I_i}(u,v,w) \\ &= max\{\nu_{A_i}(u), \nu_{B_i}(v), \nu_{C_i}(w)\}. \end{aligned} \tag{6}$$

Let $A' = (\mu_{A'}, \nu_{A'})$ and $B' = (\mu_{B'}, \nu_{B'})$ represent intuitionistic fuzzy

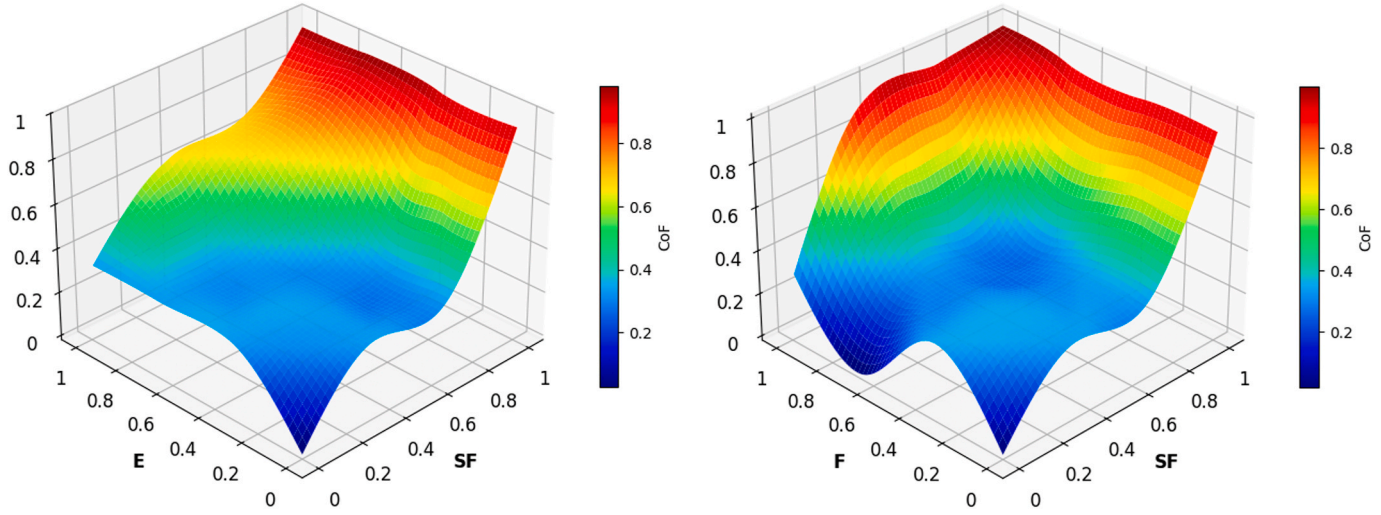


Fig. 5. Control surface of the effect of two input factors (SF, F, or E) on CoF while other factors are constant.

Probability	RISK				
0.99	0.099	0.297	0.495	0.693	0.9801
0.70	0.07	0.210	0.35	0.490	0.693
0.50	0.05	0.15	0.25	0.35	0.495
0.30	0.03	0.09	0.15	0.21	0.297
0.10	0.01	0.03	0.05	0.07	0.099
Consequence	0.10	0.30	0.50	0.70	0.99
Low Risk		Medium Risk		High Risk	

Fig. 6. Risk matrix of probability and consequence.

input observations. The output inferred from the *i*th rule using the generalized modus ponens is obtained through the sup-min composition for membership and the inf-max composition for non-membership, expressed by Eq. (7) as follows:

$$\mu_{C_i'}(w) = \sup_{u,v} \min\{\mu_{A'}(u), \mu_{B'}(v), \mu_{I_i}(u,v,w)\}, \nu_{C_i'}(w) = \inf_{u,v} \max\{\nu_{A'}(u), \nu_{B'}(v), \nu_{I_i}(u,v,w)\}. \quad (7)$$

Finally, the aggregated intuitionistic fuzzy output $C' = (\mu_{C'}, \nu_{C'})$ is obtained by combining the contributions of all m rules, given by Eq. (8) as follows:

$$\mu_{C'}(w) = \max_{i=1,\ldots,m} \mu_{C_i'}(w), \nu_{C'}(w) = \min_{i=1,\ldots,m} \nu_{C_i'}(w). \quad (8)$$

The IFIS evaluates all possible combinations of the five input parameters for the PoF, resulting in 3125 distinct intuitionistic fuzzy implications, and the three input parameters for the consequence of failure, resulting in 125 distinct intuitionistic fuzzy implications, which are implemented using the Mamdani approach. A representative set of IFIS rules expressed in If-Then format is presented as follows:

Rule 0001: If C is VL and TPD is VL and S is VL and IO is VL and G is VL, then PoF is VL.

Rule 0002: If C is VL and TPD is VL and S is VL and IO is VL and G is L, then PoF is L.

Rule 1500: If C is M and TPD is L and S is VH and IO is VH and G is VH, then PoF is VH.

Rule 3124: If C is VH and TPD is VH and S is VH and IO is VH and G is H, then PoF is VH.

Rule 3125: If C is VH and TPD is VH and S is VH and IO is VH and G is VH, then PoF is VH.

The input parameters are mapped to a value range to calculate the PoF and CoF values using intuitionistic fuzzy rules and the constructed inference system. Figs. 4 and 5 illustrate the partial mapping of the correlation input factors on output PoF and CoF, while other factors are held constant. From the mapping, it can be seen that as input increases, output increases as well, consistent with actual logical reasoning. The mapping is a high-dimensional correlation that is difficult to describe using explicit formulas (Yin et al., 2021).

3.2.3. *Intuitionistic defuzzification*

The defuzzification process transforms intuitionistic fuzzy output sets into crisp numerical values for further analysis. The centroid method is the most widely used approach to intuitionistic defuzzification. In this study, we propose a defuzzification operator z^{Θ} that is defined by Eq. (9) as follows (Velu and Paulraj, 2025):

$$z^{\Theta} = \frac{\int_z z * \left[\Theta * \mu_{C_k}(z) + (1-\Theta) * \left(1 - \nu_{C_k}(z)\right)\right] dz}{\int_z \left[\Theta * \mu_{C_k}(z) + (1-\Theta) * \left(1 - \nu_{C_k}(z)\right)\right] dz}. \quad (9)$$

The balance between membership and non-membership information in the intuitionistic defuzzification process is influenced by the parameter $\Theta \in [0,1]$. Larger Θ values reduce the influence of uncertainty and produce more decisive outputs, whereas smaller Θ values are more sensitive to uncertainty. If $\Theta = 1$, the classical center-of-gravity (COG) defuzzification is obtained.

Risk is evaluated through a systematic application of the intuitionistic fuzzy method, which addresses uncertainty and yields crisp values for the PoF and CoF. The proposed method provides a structured system for evaluating the PoF and CoF, the two fundamental aspects of risk assessment, in accordance with Eq. (10). Subsequently, a probability–consequence matrix is applied to the calculated risk values to rank and prioritize inspection and maintenance activities. The developed risk matrix, modified from (Sadeghi et al., 2025), serves as the fundamental framework for risk prioritization throughout the study (Fig. 6).

$$Risk = PoF * CoF \quad (10)$$

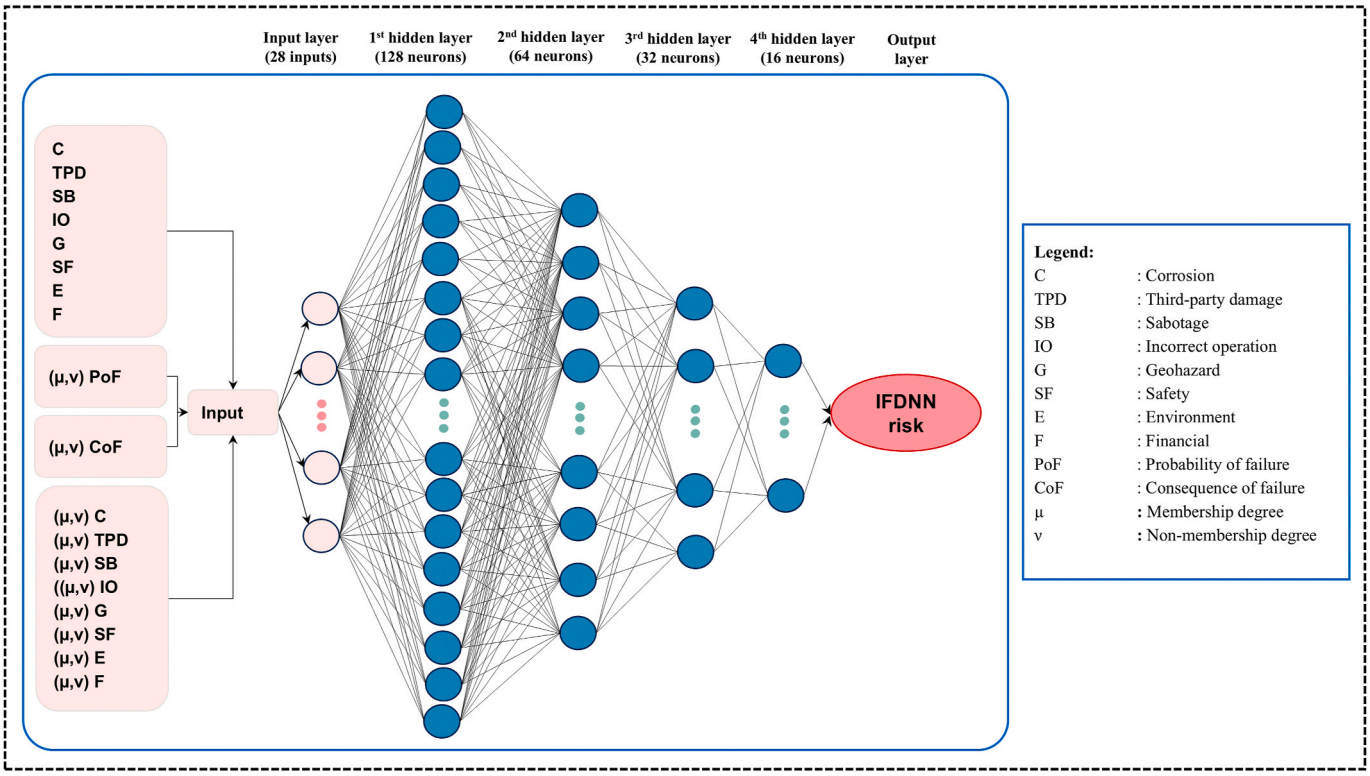


Fig. 7. The proposed structure of the selected IFDNN model.

3.3. Stage III: intuitionistic fuzzy deep neural network risk assessment model

The proposed model integrates intuitionistic fuzzy sets with a deep neural network (IFDNN) to predict failure risk. The robust IFDNN approach enables processing large-scale data using deep learning techniques. The structural design was built sequentially using IFS tools within a DNN with multiple hidden layers. This structure is suitable for solving high-nonlinearity problems and for faster convergence (Talpur et al., 2023). The model was developed with a Python-based deep learning framework (TensorFlow–Keras). The proposed structure of the selected IFDNN model is shown in Fig. 7. The input features are 8 main factors from the traditional model, and their values are converted into intuitionistic fuzzy sets, represented by membership and non-membership degrees. The deep neural network then generates the output of the sequential IFDNN model.

The dataset used in this study comprises approximately 1750 km of onshore upstream oil and gas pipelines across 15 operational fields. The pipeline primarily transports produced fluids from the well to processing facilities, produced water from the processing plant to injection wells, and crude oil from processing facilities to storage tanks and refinery units. For analysis, the pipeline network was divided into 19,405 subsegments, which constituted the dataset used to develop and evaluate the proposed model. Pipeline diameters range from 4 to 40 inches, while asset ages span from less than 10 years to more than 50 years, representing a diverse mix of new and aging infrastructure. Some networks, which have operated for several decades and are located near populated areas, exhibit significant corrosion, thereby posing significant integrity and safety concerns. The diversity of asset ages, diameters, degradation conditions, service types, and geographic locations provides a representative dataset for evaluating the generalizability of the proposed framework.

DNN is a deep learning framework in which neuron pairs simultaneously establish connections across multiple layers. The DNN prediction model's structure is beneficial and can therefore be easily scaled across multiple datasets and constraints (Guang et al., 2024). The DNN model was developed using the sequential features of an intuitionistic fuzzy input, four fully connected hidden layers, and a single intuitionistic fuzzy risk output node. The dataset comprises 19,405 pipeline sub-segments characterized by 28 features, including the main risk factors, the intuitionistic fuzzy membership and non-membership degrees of the main risk factors, and the corresponding intuitionistic fuzzy representations of the probability of failure and the consequences of failure. The resulting intuitionistic fuzzy risk values were used as the target labels for model training and evaluation. The dataset was randomly partitioned into training (70%), testing (20%), and validation (10%) subsets. The validation dataset was excluded from model training and used solely for independent performance evaluation, ensuring an unbiased assessment of the model's predictive capability and generalizability.

Several deep neural network models were evaluated to identify the optimal models for the study's objective. The multilayer perceptron (MLP), widely used and with a strong theoretical foundation, was chosen for its effectiveness in the non-linear modeling of complex datasets. A feedforward DNN was selected as the framework for risk prediction, given the complex, non-linear pipeline risk assessment, with forward propagation for output estimation and backpropagation for weight optimization during training. The deep neural network was designed using a more practical, iterative approach that included, as input layers (or input nodes), a series of hidden layers (for hierarchical extraction) and an output layer (for risk prediction).

The model is evaluated to identify the optimal architecture and performance. A network architecture comprising 128, 64, 32, and 16 neurons across the hidden layers demonstrates superior predictive performance. Several activation functions were tested in the hidden layers to identify the best configuration. Comparing the input activation function of the rectified linear unit (ReLU) and sigmoid linear unit (SiLU), the Gaussian error linear unit (GELU), and a linear activation function in the output layer, the GELU achieves the lowest MAE, MSE, RMSE, and the highest R^2. Training was performed using the Adam

Table 2
Selected activation function in a two-layer network.

Network performance				Activation functions			Number of runs
R^2	MAE	MSE	RMSE	Input - hidden	Hidden - hidden	Hidden - output	
0.99265	0.009632	0.000266	0.016297	ReLU	ReLU	Linear	1
0.99100	0.007376	0.000163	0.012777	SiLU	SiLU	Linear	2
0.99275	**0.006401**	**0.000131**	**0.011466**	**GELU**	**GELU**	**Linear**	3
0.99095	0.007030	0.000164	0.012810	ReLU	GELU	Linear	4
0.99204	0.006659	0.000144	0.012011	GELU	ReLU	Linear	5

Table 3
Neural network performance across different hidden layers.

R^2	MAE	MSE	RMSE	Activation functions (left to right)	Number of layers
0.99275	0.006401	0.000131	0.011466	GELU - GELU - Linear	4 (two hidden layers)
0.99266	0.006533	0.000133	0.011534	GELU - GELU - GELU - Linear	5 (three hidden layers)
0.99443	**0.006791**	**0.000101**	**0.010046**	**GELU - GELU - GELU - GELU - Linear**	**6 (four hidden layers)**
0.99251	0.005269	0.000136	0.011655	GELU - GELU - GELU - GELU – GELU - Linear	7 (five hidden layers)

Table 4
Performance of the two selected layers for training, testing, validation, and all data.

DNN layer types and evaluation criteria	Training data	Testing data	Validation data	All data
R^2	0.99455	0.99458	0.99330	**0.99443**
MAE	0.006761	0.006801	0.006981	**0.006791**
MSE	0.000100	0.000094	0.000122	**0.000101**
RMSE	0.009990	0.009703	0.011064	**0.010046**

Table 5
Pipeline information data.

Parameter	Segment 1	Segment 2	Segment 3
Diameter (inch)	16	16	6
Nominal thickness (mm)	6.35	9.53	7.11
Segment length (m)	500	500	5000
Service life (years)	44	30	45
Operating pressure (psig)	260	470	450
Operating temperature (°F)	200	180	150
Service fluid	Crude oil – production	Produced water	Crude oil - transmission
Construction material	CS API 5L Gr.B PSL 2	CS API 5L Gr.B PSL 2	CS API 5L Gr.B PSL 2
Pipeline external condition	Aboveground	Aboveground	Aboveground

optimizer, which provides efficient convergence in Python-based deep learning frameworks. Tables 2 and 3 show that uniform activation functions outperform hybrid configurations in terms of performance, although increasing the network depth slightly diminishes its effectiveness. Finally, we employed a DNN with four hidden layers and the GELU activation function.

Deep neural networks can learn complex data by selecting an appropriate number of hidden layers. A small number of neurons selected by the DNN prevents it from learning the most available patterns. It also revealed that many neurons contributed to pattern preservation, indicating that the DNN could not learn to recognize its intrinsic properties. The network performance of the four hidden layers outperforms the others, as shown in Table 4. Next, to improve the model, the DNN was trained to connect the input and output layers, resulting in valid outputs. Therefore, the default input and output values for this data were already entered into the DNN structure in the previous IFS models. Risk assessment using the model is performed after a thorough training process, and the model's efficiency is validated on 10% of the dataset. Data were not used for training and testing.

3.4. Method of performance evaluation

The risk prediction accuracy was measured to obtain the expected performance of deep learning models. The evaluation criteria are R^2, MAE, MSE, and RMSE. The R^2 ranges from 0 to 1 and indicates better predictive performance when it is close to 1. Other metrics indicate higher predictive performance when the value is close to 0. These measures are described using the following mathematical expressions given by Eqs. (11)–(14) as follows:

$$R^2 = \frac{\sum_{i=1}^{N} \left(\hat{y}_j - \bar{y}\right)^2}{\sum_{i=1}^{N} \left(y_j - \bar{y}\right)^2} \tag{11}$$

Table 7
Factors determining the PoF and CoF according to the IFS models.

Segments	Risk aspects		Intuitionistic fuzzy risk	
Segment 1	Probability	0.7999	0.2969	Medium
	Consequence	0.3712		
Segment 2	Probability	0.2897	0.2318	Medium
	Consequence	0.8000		
Segment 3	Probability	0.7999	0.6399	High
	Consequence	0.7999		

Table 6
Factors determining the PoF and CoF according to traditional models.

Parameters	Probability factors					Consequence factors		
Segments	Corrosion	TPD	Sabotage	IO	Geohazard	Safety	Financial	Environment
Segment 1	5	1	1	1	1	1	3	1
Segment 2	3	1	1	1	1	1	4	1
Segment 3	5	1	1	1	1	2	1	5

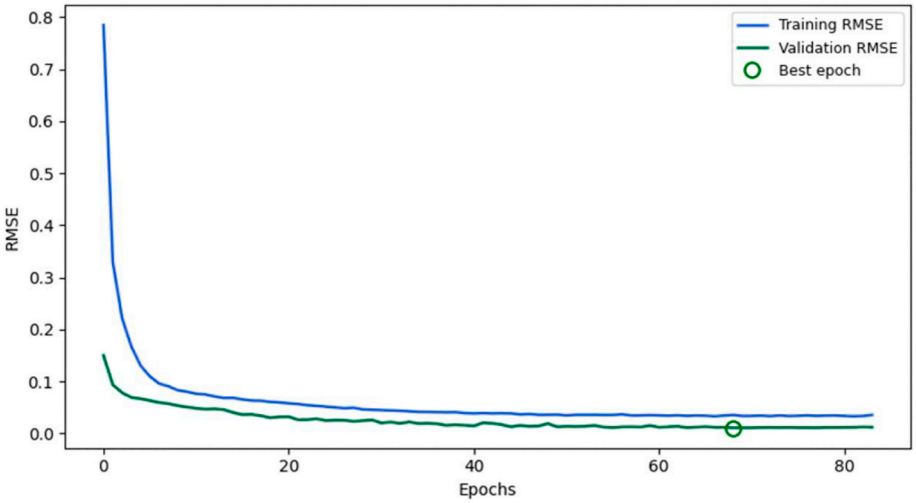


Fig. 8. The training process of intuitionistic fuzzy deep neural networks.

$$\mathrm{MAE} = \frac{1}{N}\sum_{j=1}^{N}\left|y_j - \hat{y}_j\right| \tag{12}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{j=1}^{N}\left(y_j - \hat{y}_j\right)^2 \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_j - \hat{y}_j\right)^2} \tag{14}$$

where $\overline{y}$, y_j, $\hat{y}_j$ are the average values of the target data, jth target value, and jth predicted value, respectively, and N is the number of data samples.

3.5. Case study and model implementation

The proposed IFDNN risk assessment model was verified by a real-world case study across pipeline types: oil production, water injection, and oil transmission. These pipeline categories were selected to reflect typical configurations encountered in upstream onshore pipeline facilities and to demonstrate the general applicability and feasibility of the proposed risk assessment framework across different operational functions and risk profiles. Fig. 10 shows an illustrative schematic of the pipeline network, and the key variables and characteristics of the pipeline segments considered in the analysis are summarized in Table 5.

The pipeline risk assessment was conducted using a comprehensive framework and systematic process. These steps include pipeline segmentation, estimation of threat exposure, evaluation of mitigation measures, assessment of pipe wall resistance, calculation of the PoF and the CoF, and risk estimation derived from integrating PoF and CoF. The risk criteria are examined at the sub-segment level, with breaks made based on different operating conditions and differences in construction and environmental characteristics along pipeline routes, which may expose them to varying levels of risk. Accordingly, risk is evaluated at the sub-segment level, with new sub-segments defined when significant changes occur in construction practices, operating history, or surrounding conditions. Relevant threats are identified in each sub-segment, and baseline exposure is assessed, with a clear distinction between time-dependent and event-driven, time-independent degradation mechanisms.

In risk assessment, every plausible failure mechanism has three main elements: exposure, mitigation, and resistance, and they should be considered in any risk evaluation mechanism. The failure mechanism: exposure is an attack with no mitigation, and failure ensues. The planned risk reduction of threat exposure can be achieved through engineering controls, inspection and monitoring activities, or operational controls. In line with the Muhlbauer approach, resistance is measured according to the structural capacity in place, the presence or absence of existing or potential flaws, degradation rates, inspection frequency, and inspection effectiveness. The PoF is assessed by integrating factors such as exposure, the effectiveness of mitigation strategies, and resistance into a unified likelihood value for each segment. The primary considerations of failure consequences are safety, environmental, and financial. The combined PoF and CoF values yield risk rankings that support inspection prioritization, mitigation planning, and risk-informed pipeline integrity management.

Intuitionistic fuzzy risk assessment is performed according to Fig. 2, in which all failure factors for the PoF and CoF are initially expressed as linguistic variables to explicitly capture the epistemic uncertainty and imprecision inherent in expert judgments and field data. A fuzzification process transforms linguistic variables into triangular intuitionistic fuzzy numbers (TIFNs), thereby facilitating the simultaneous modeling of membership, non-membership, and hesitation components to capture uncertainty better. Conversion from input to output is conducted using an inference engine based on a fuzzy rule base. Subsequently, the intuitionistic fuzzy values are defuzzified to obtain crisp PoF and CoF scores that preserve the underlying uncertainty information. Finally, the integrated PoF and CoF values are combined to compute an overall risk index, which is classified into low, medium, and high-risk levels to support practical RBI planning and informed decision-making.

Furthermore, the proposed intuitionistic fuzzy deep neural network (IFDNN)–based risk assessment framework is implemented by transforming intuitionistic fuzzy membership and non-membership degrees into structured inputs for a deep neural network (DNN), thereby enabling the model to account for uncertainty in the input data explicitly. The DNN produces a continuous risk score that reflects non-linear interactions among contributing risk factors, which is subsequently mapped to separate risk levels using the predefined risk matrix illustrated in Fig. 6. The risk prediction outcomes generated by the traditional risk assessment method, intuitionistic fuzzy method, and IFDNN-based method are systematically compared and analyzed to evaluate the effectiveness and reliability of the proposed approach, allowing for a comprehensive assessment of performance improvements in terms of prediction consistency, discrimination capability, and robustness under varying operational conditions.

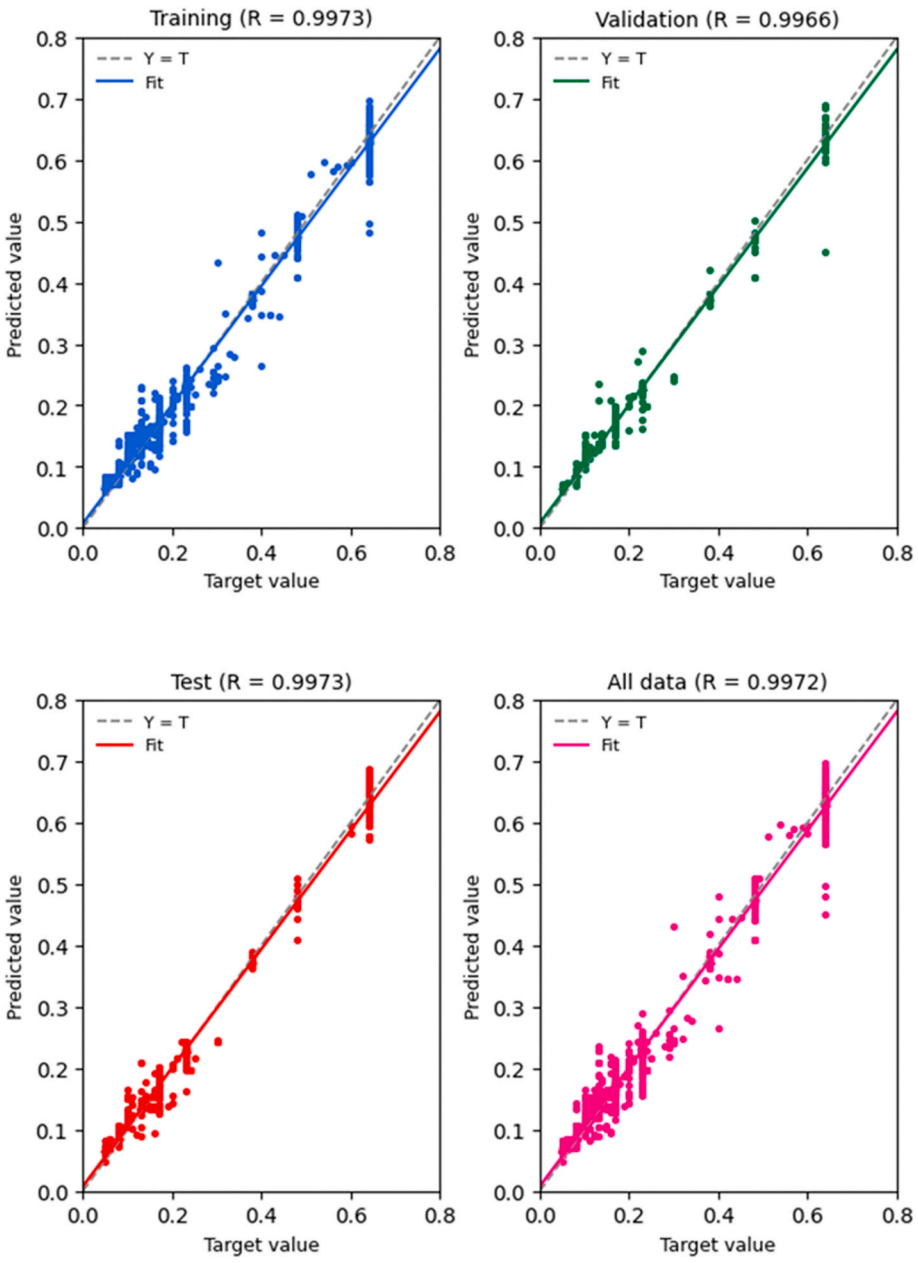


Fig. 9. Optimal IFDNN model outputs based on actual datasets of training, testing, validation, and all data.

4. Results and discussion

Table 6 presents the PoF and CoF for each segment, as determined using the Muhlbauer risk assessment model and API RP 581. As the main factors, these figures serve as the foundational quantitative data for risk assessments in the proposed model and indicate the comparative probabilities and consequences of potential failure scenarios across pipeline categories. According to the main factors summarized in Table 6, the corresponding intuitionistic fuzzy risk assessment outcomes are reported in Table 7. The consistency between the baseline Muhlbauer-based assessments and intuitionistic fuzzy results demonstrates the validity of the proposed approach in capturing uncertainty while preserving the pipeline system's underlying risk characteristics.

Fig. 8 shows the DNN model's performance during training and validation, measured by RMSE, with the best validation performance indicating the epoch at which optimal generalization is achieved. The convergence behavior of the RMSE curves indicates stable learning, and the optimal epoch is selected according to the minimum validation RMSE to ensure proper generalization and avoid overfitting. Strong predictive performance is demonstrated across all evaluation measures across all datasets. Fig. 9 compares the outputs to the target and predicted risk values of the optimal DNN model across training, testing, and validation datasets. The model prediction was represented by a fitted regression line, with the diagonal line representing the ideal target. Finally, the trained model demonstrated excellent performance.

The enhancement of the proposed model, as shown in Table 8, is compared with that of traditional models. In traditional models, combining different inputs can lead to the same risk, even when different risks lead to errors in analysis and inefficient use of resources and costs. This weakness stems from the inability to differentiate among risk factors and consider the more important variables. In contrast, the proposed risk model accounts for all input variables and interdependencies, enabling risk differentiation across pipeline segments to support decision-making prioritization. The boundaries between different risk levels in the traditional model span a wider range than in the intuitionistic fuzzy deep neural network method, so risk assessment

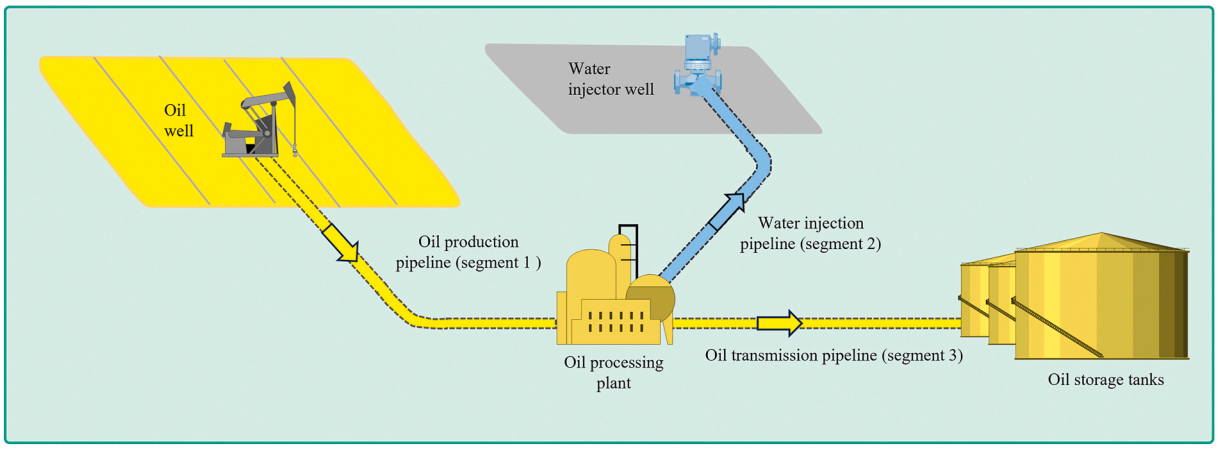


Fig. 10. Schematic of the pipeline segment network.

Table 8
Comparative performance evaluation of traditional, IFS, and IFDNN models.

Segments	Traditional		IFS		IFDNN	
Segment 1	15	High	0.2969	Medium	0.2428	Medium
Segment 2	12	Medium	0.2318	Medium	0.2209	Medium
Segment 3	25	High	0.6399	High	0.5759	High

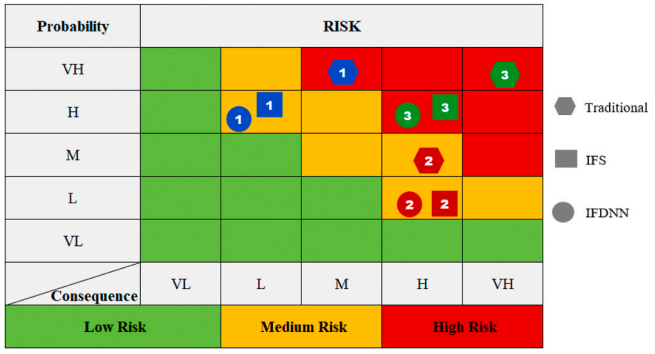


Fig. 11. Risk plot for traditional, IFS, and IFDNN models.

in the latter will be less accurate. The accuracy of the proposed model can be explained by the fact that the correlation between inputs and outputs is described in terms of linguistic variables, terms, and conditions, as well as correlations among input parameters that contribute to risk prediction.

The performance of risk prediction using a hybrid intuitionistic fuzzy logic with deep neural networks has significantly improved over traditional models, providing more precise and reliable risk predictions, even for complex, uncertain, and highly non-linear data. The key difference between the traditional and proposed models is the identification and assessment of the relevance of each sub-factor and main factor to pipeline risk. As indicated in Table 8 for Segment 1, the conventional model is classified as high risk, whereas the IFDNN model is classified as medium risk based on the analysis of critical factors. The new model demonstrates that the hybrid framework delivers better performance. Additionally, this combination enhances feature representation and generalization, thereby improving predictive accuracy and robustness, particularly when addressing the variability and uncertainty inherent to real-world pipeline integrity assessments. Improving the prediction of high-risk to medium (segment 1) will affect inspection interval activity, ultimately reducing operational expenditure without sacrificing operational safety.

Figs. 11 and 12 present the risk categorization and prioritization for the three pipeline segments using traditional, IFS, and IFDNN models. The traditional model produces the highest risk estimates, classifying Segments 1 and 3 as high risk and Segment 2 as medium risk. In contrast, the proposed models reduce the number of risk categories and the risk score by incorporating uncertainty and non-linear relationships among risk factors. Despite these reductions, the risk prioritization remains unchanged, with Segment 3 > Segment 1 > Segment 2, demonstrating the robustness of the proposed model in identifying critical pipeline

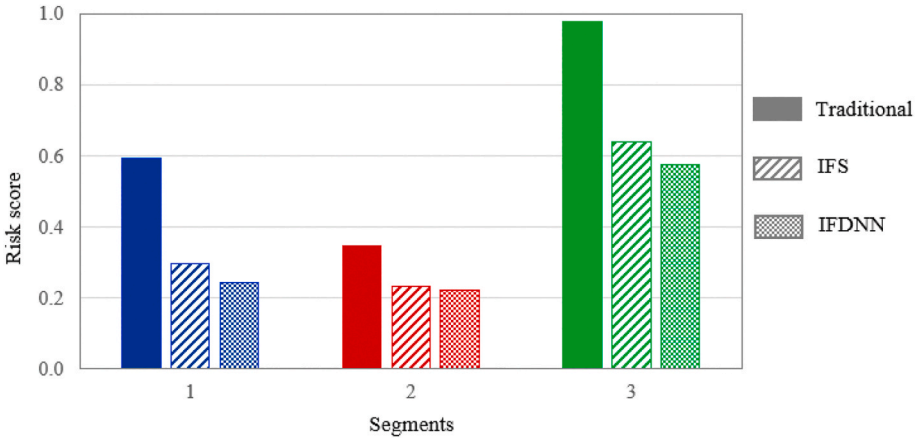


Fig. 12. Comparative analysis of risk prioritization results.

segments. Segment 3 consistently exhibits the highest risk because the pipeline is relatively aged and exposed to a more corrosive environment, increasing the probability of failure. In addition, Segment 3 has the most severe consequences in the event of failure due to its proximity to public and environmentally sensitive areas, whereas an oil spill could cause substantial environmental and financial impacts. Corrosion is identified as a dominant contributor to failure probability, as reflected in the available repository, inspection interval, examination, accident, and maintenance records. These findings indicate that risk mitigation efforts should prioritize Segment 3, with particular attention given to corrosion-related sub-factors and the key driver of environmental consequence to support more effective pipeline integrity management.

The risk analysis results indicate that corrosion is the dominant factor contributing to the highest pipeline risk. This finding suggests that PIM strategies should focus on proactive management and prediction of corrosion-related damage mechanisms. The asset integrity team can take priority action to strengthen the effectiveness of the PIM program. First, in the periodic inspection plan, priority is given to a detailed internal inspection of the pipeline using high-resolution methods to identify internal and external corrosion. Second, a maintenance program should be enhanced to mitigate corrosion through effective chemical injection and coating application for internal and external corrosion, respectively. The selected chemical treatments should be appropriate for the prevailing damage mechanism, such as microbiological influence corrosion (MIC), carbon dioxide (CO2) corrosion, or other corrosion-related threats. Third, a repair or replacement program should be implemented for pipeline sections that fail to satisfy fit-for-service assessment criteria. In addition, the resource should be reallocated to high-risk pipeline segments to optimize risk-based return on investment.

The proposed model enhances and refines the existing RBI and PIM standard process. This can transform a fixed-cycle inspection plan into a risk-based, predictive maintenance program. The traditional RBI often relies on historical data and expert judgment when prioritizing inspections. The proposed method provides more robust scientific decision support by quantifying uncertainty and accounting for non-linear relationships among risk factors. The implementation of an integrity management strategy based on precise risk ranking is expected to provide significant benefits in reducing the risk of failure, optimizing the maintenance program, and extending asset service life. By focusing on the risk drivers, minimizing unnecessary inspection of low-risk areas, and ensuring adequate attention to high-risk segments, it can more effectively prevent major incidents such as leaks or ruptures, thereby mitigating safety, financial, and environmental consequences. User training is required to ensure proper interpretation and utilization of input requirements, output interpretation, and limitations. A phased deployment approach involving data validation, system integration, pilot testing, and user capacity building is recommended to support successful industrial adoption.

Although the proposed model shows strong potential to improve the prediction of pipeline operational risk by effectively handling uncertainty and complex non-linear data, several limitations should be acknowledged in future work and recommendations. The current study is based on onshore pipeline data, and its applicability to offshore pipeline systems with different operating and environmental conditions remains to be validated. Furthermore, the proposed model primarily evaluates individual risk factors and does not explicitly capture the complex interactions and dependencies among multiple threats, such as corrosion, mechanical damage, and material defects, which may collectively influence the probability of failure. In addition, while intuitionistic fuzzy sets enhance the representation of uncertainty, they may not capture all forms of complex uncertainty inherent in dynamic pipeline operations. For instance, time-dependent damage factors evolve continuously, requiring real-time monitoring to enable accurate, adaptive risk assessment. Therefore, to support dynamic risk prediction based on real-time data and generate dynamic risk estimates, it is necessary to integrate an appropriate deep learning model to deliver accurate and efficient real-time predictions.

5. Conclusion

This study developed a novel framework integrating intuitionistic fuzzy sets and deep neural networks to enhance the risk assessment of oil and gas pipelines. The proposed model effectively addresses the limitation of a traditional approach that relies on a single method, offering a better solution for managing uncertainty and handling complex risk data. Risk analysis was carried out based on 31 sub-factors and 8 main factors, which were categorized into probability of failure and consequence of failure. The main factors derived from traditional models, combined with intuitionistic fuzzy sets, were used as inputs to deep neural networks to predict the risk. The dataset used in this study was obtained from a pipeline risk assessment conducted on onshore pipeline oil and gas production.

The findings demonstrate that intuitionistic fuzzy deep neural network models provide robust frameworks for managing uncertainty and complex, non-linear data, thereby enhancing the accuracy of risk predictions. The proposed hybrid model outperforms traditional methods, achieving higher accuracy in predicting risk levels. The boundaries between different risk levels in the traditional model span a wider range than in the intuitionistic fuzzy deep neural network method, so risk assessment in the latter will be less accurate. The intuitionistic fuzzy deep neural network method's higher accuracy can be explained by modeling the relationship between input and output as a linguistic variable, which is more realistic. This improvement is largely attributed to its ability to incorporate the relative significance of multiple influencing factors, which are often overlooked in traditional methods. The improved accuracy of the proposed model helps prevent serious accidents while maximizing pipeline integrity at the lowest possible cost. By enhancing predictive capability and decision support, it reduces the likelihood of major incidents while supporting more cost-efficient pipeline integrity management strategies.

Future studies should extend and validate the proposed framework using offshore pipeline datasets, where more complex environmental and operational conditions may introduce additional degradation mechanisms and sources of uncertainty. Further research should also investigate the effects of threat interaction and dependencies among multiple pipeline threats on failure probability and explore advanced fuzzy set generalizations to enhance uncertainty quantification. In addition, integrating real-time monitoring data with advanced deep learning architectures facilitates dynamic risk assessment and strengthens the predictive and adaptive capabilities of the framework for pipeline integrity management.

CRediT authorship contribution statement

Tri Wahono: Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Imam Mukhlash:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Endah R.M. Putri:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Agung Purniawan:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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